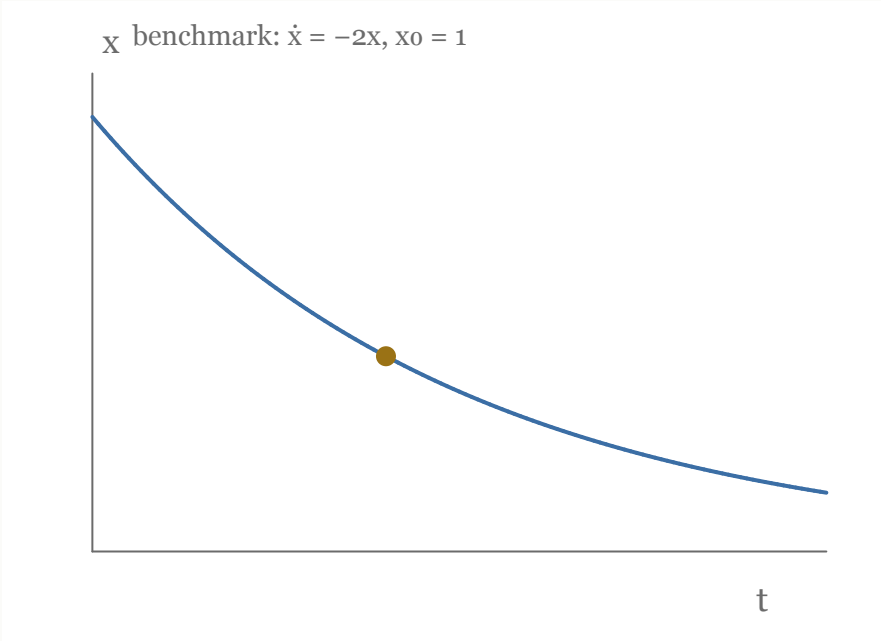
ODE and Simulation

How does a simulator advance the world?

$$\dot{x} = f(x, t), \qquad x(t_0) = x_0$$

- `x` state column
- `f` rate of change
- `t` time; to is the start
- x_0 known initial state

A model gives a rate. A simulator produces a future state.



OBJECTS

ODE and Simulation

A trajectory and its stored samples

$$t_k = t_0 + kh, \qquad x_k \approx x(t_k)$$

`h, k` step size and sample index

`x(t), xₖ` exact state and numerical sample

Rate model

↓ is sampled by

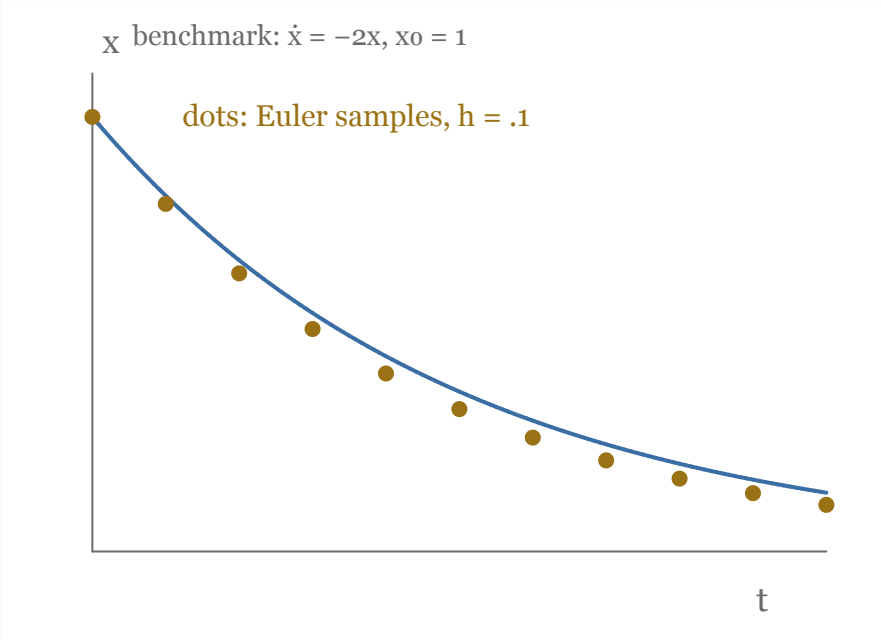
Time-stepping method

↓ produces

Numerical trajectory

↳ Solver check: exact / refined solution

↳ Model check: measurements



DEFINITION

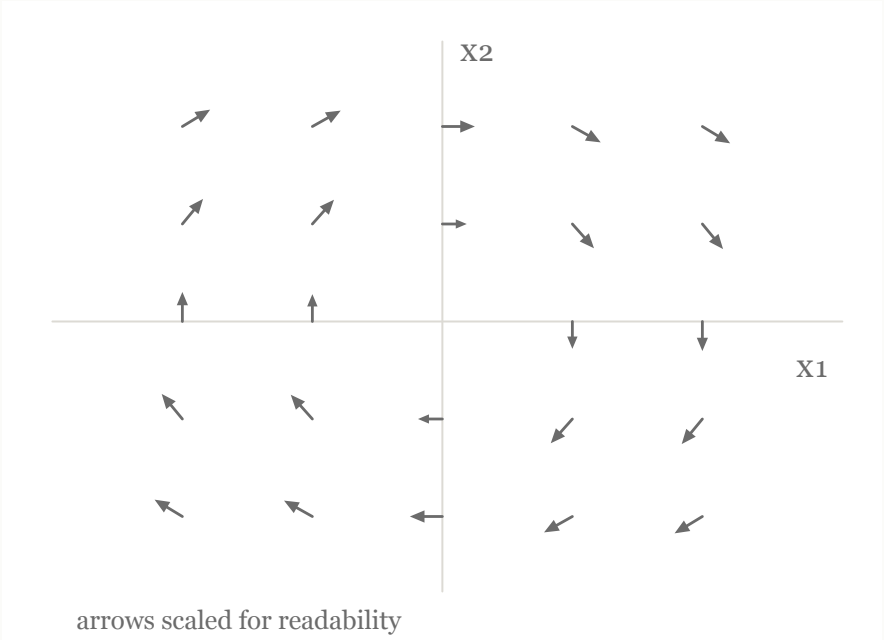
Velocity Fields

An arrow gives a rate, not a destination

$$\dot{x} = f(x, t) \in \mathbb{R}^d$$

- `x` d-component state column
- `f(x,t)` d-component state rate
- `t` time held fixed in a field picture

Illustration: $x=(q,v)^T$ and $f=(v,-\sin q)^T$. Arrow lengths are scaled.



arrows scaled for readability

ASSUMPTIONS

Velocity Fields

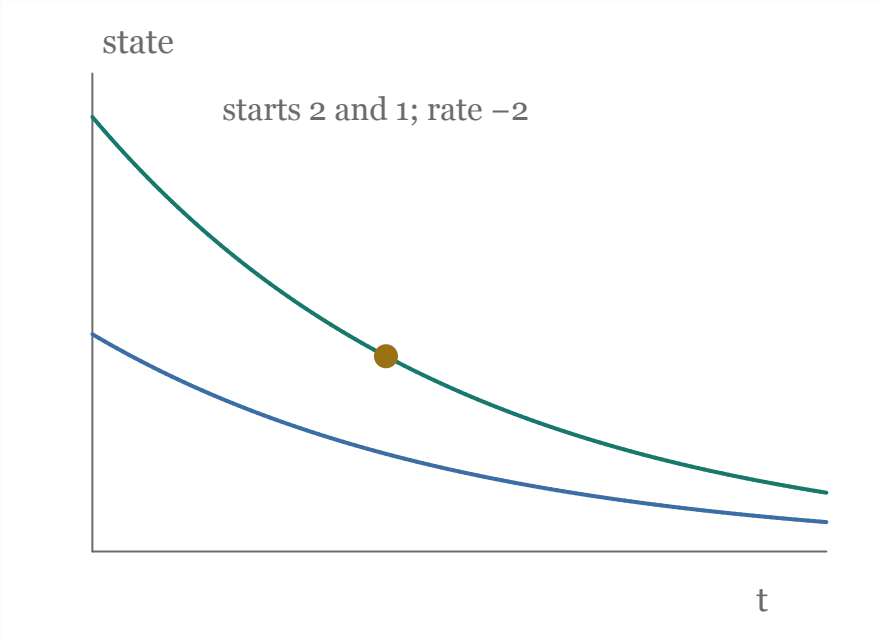
One field, different initial states

$$\dot{x} = -2x, \quad x(0) = 1 \text{ or } 2$$

`x(0)` two possible starts

`-2x` the same rate model for both

An initial-value problem combines the rule with its starting state.



WORKED EXAMPLE

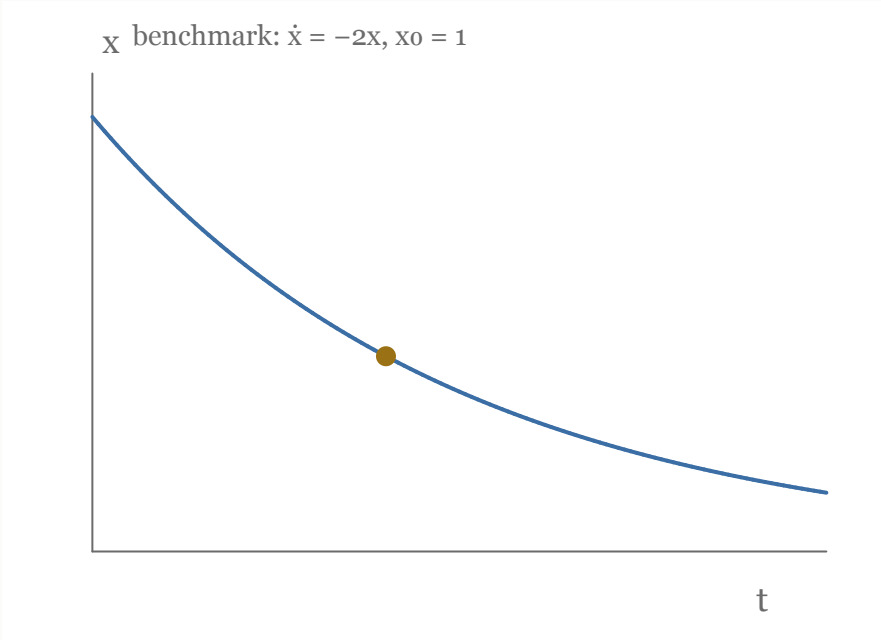
Velocity Fields

Verify an exact decay solution

$$x(t) = e^{-2t} \quad \Longrightarrow \quad \dot{x} = -2x$$

- `x(0)` one
- `-2` decay rate in inverse time
- `t` elapsed time

> Worked example: x(0.1) = 0.8187307531.



YOUR TURN

Velocity Fields

Your turn: a changed decay rate

$$\dot{x} = -3x, \qquad x(0) = 2$$

- `x` scalar state
- `-3` fixed decay rate
- `2` initial value

Which trajectory satisfies BOTH the rate and the start?

A. exp(−3t)

B. 2 exp(−3t)

C. 2 exp(3t)

WHY

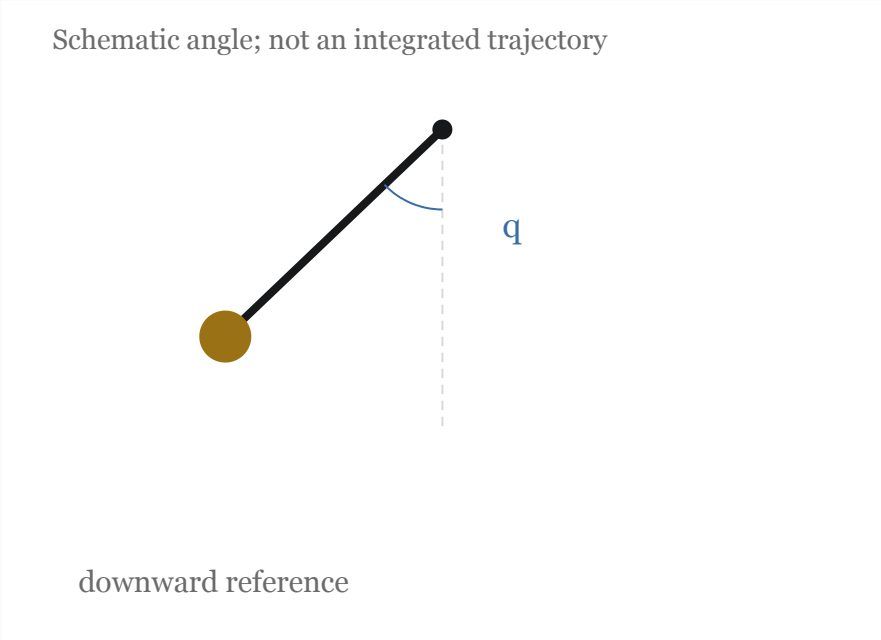
Robot State

Position alone does not predict motion

$$x = \begin{bmatrix} q \\ v \end{bmatrix}, \qquad v = \dot{q}$$

- q n joint positions
- v n joint velocities
- x 2n-component state

The same angle can occur while a joint moves in either direction.



MODEL

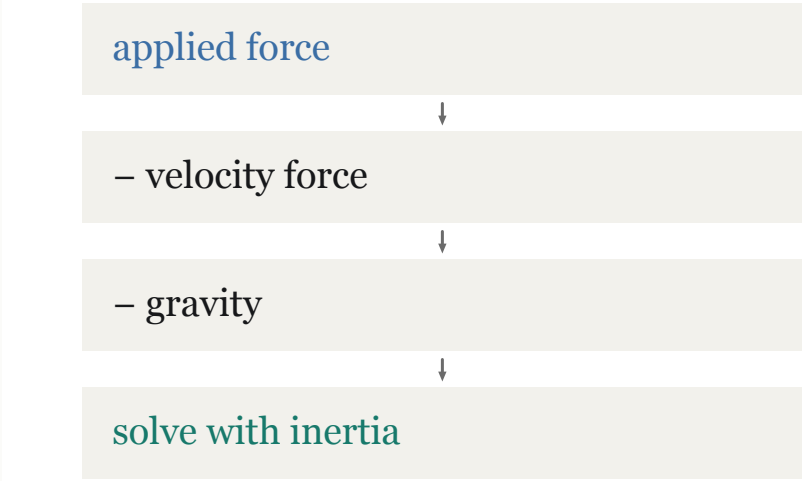
Robot State

The manipulator equation is a force balance

$$M(q)\ddot{q} + C(q, v)v + g(q) = \tau$$

`M`	n×n inertia, positive definite
`Cv`	velocity-dependent generalized forces
`g, τ`	gravity and applied generalized forces
`q,v`	positions and velocities

The model balances inertial, velocity-dependent and gravitational terms.



DERIVATION

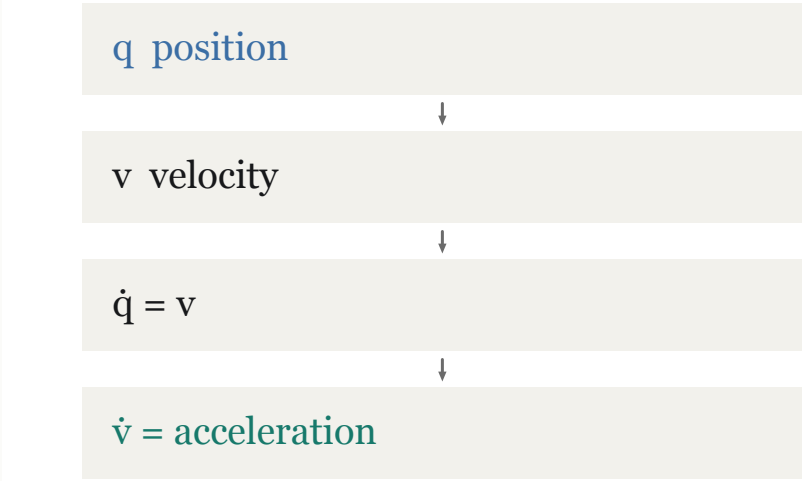
Robot State

Rewrite the mechanics as a first-order ODE

$$\dot{x} = \begin{bmatrix} v \\ M(q)^{-1}[\tau - C(q,v)v - g(q)] \end{bmatrix}$$

- x [q;v], a 2n-column
- τ applied generalized force
- M^{-1} exists for positive-definite inertia

Compute acceleration by solving M a = τ − Cv − g.



WORKED EXAMPLE

Robot State

A one-joint calculation

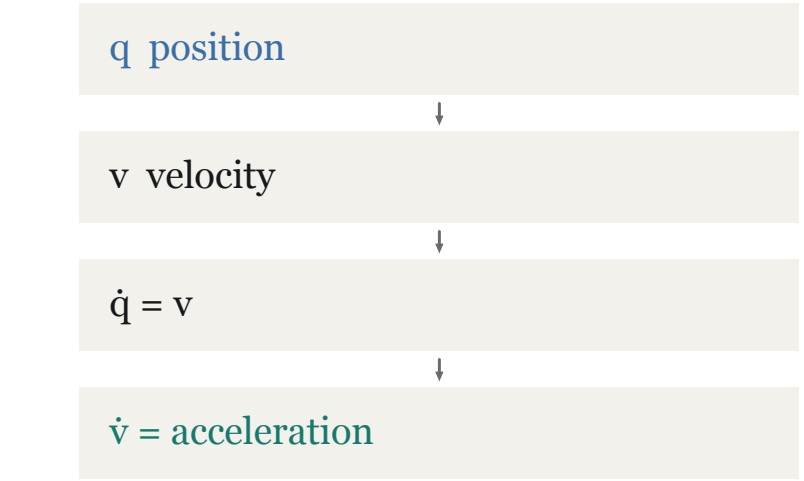
$$a = \frac{5 - 0.5(2) - 3}{2} = 0.5, \qquad \dot{x} = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

`M,C` 2 and 0.5 at this state

`g,τ` 3 and 5

`q,v` 1 and 2

> Worked example: solve acceleration, then assemble the full derivative.



YOUR TURN

Robot State

Your turn: reduce the applied force

$$M = 2,\ C = .5,\ g = 3,\ \tau = 1,\ v = 2$$
$$Ma = \tau - Cv - g, \quad \dot{x} = (v, a)^T$$

`a`	acceleration
`x=(q,v)ᵀ`	position then velocity

Which state rate follows from the force balance?

A. $(-1.5, 2)^T$

B. $(2, -3)^T$

C. $(2, -1.5)^T$

MODEL

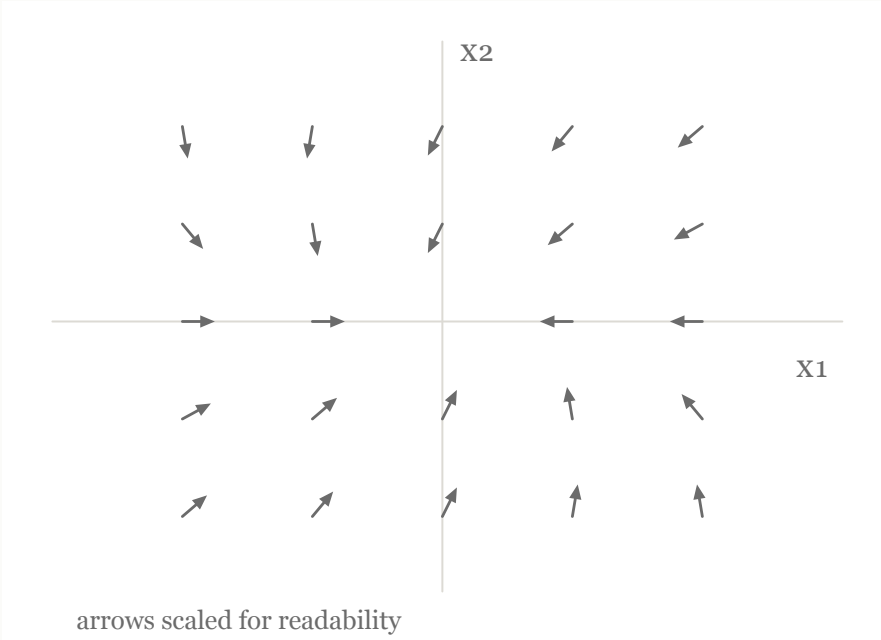
Linear Dynamics

A constant matrix gives a linear field

$$\dot{x} = Ax, \qquad A \in \mathbb{R}^{d \times d}$$

- **A** constant matrix
- **x** d-component state
- **Ax** state rate at x

Illustrated A has rows (−1,−1) and (0,−2); the field is homogeneous.



RECALL

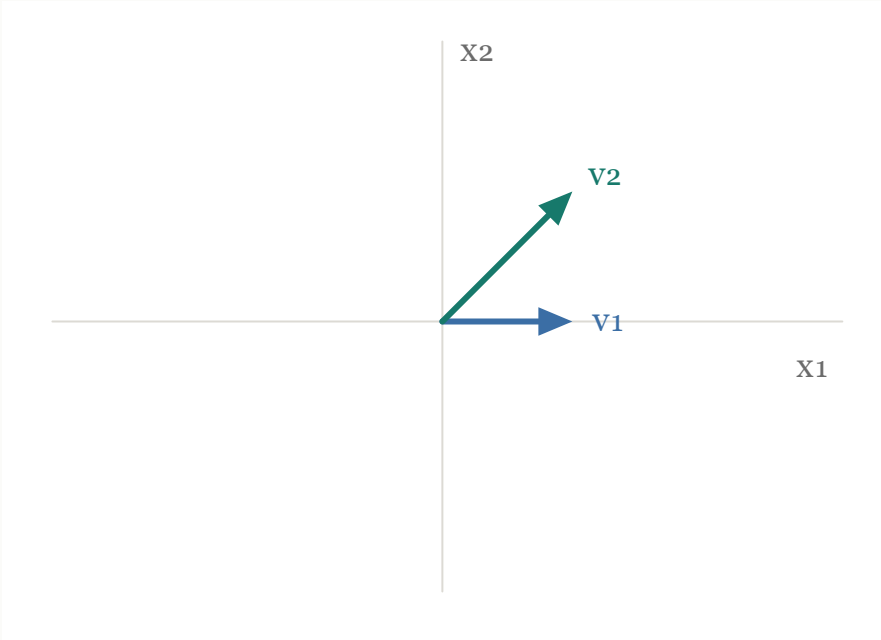
Linear Dynamics

Recall the eigenvector coordinates

$$AV = V\Lambda, \qquad x = Vc$$

- V invertible eigenvector matrix
- Λ eigenvalues in matching order
- c coordinates of x in that basis

Illustrated basis: $v_1=(1,0)^T$, $v_2=(1,1)^T$. Invertible does not mean orthogonal.



DERIVATION

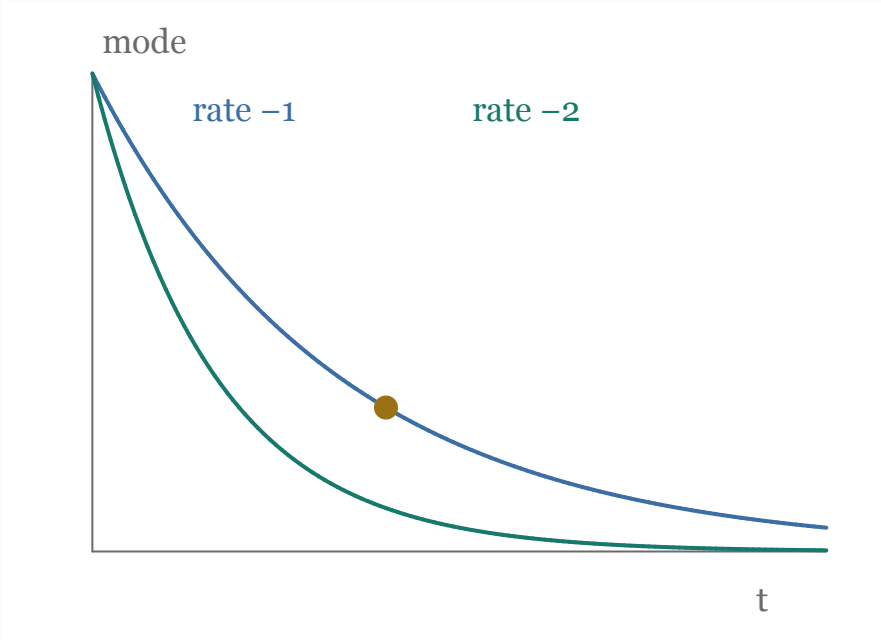
Linear Dynamics

The ODE separates into scalar modes

$$V\dot{c} = AVc = V\Lambda c \quad \Rightarrow \quad \dot{c} = \Lambda c$$

- V constant invertible basis
- c time-dependent coordinates
- Λ diagonal rate matrix

Multiply on the left by V^{-1}; each coordinate now evolves alone.



RESULT

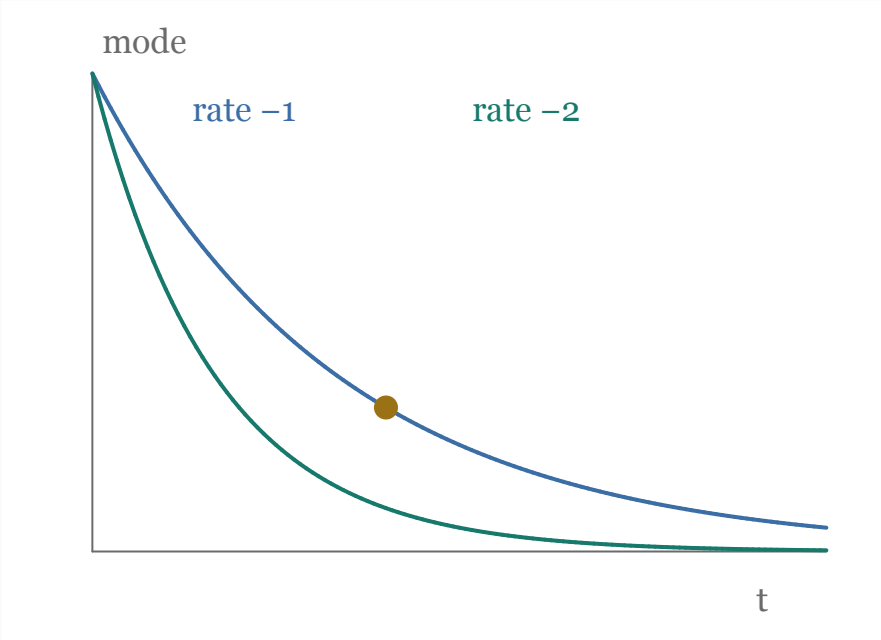
Linear Dynamics

Each eigenvalue is a continuous growth rate

$$c_i(t) = e^{\lambda_i t} c_i(0)$$

- λ_i rate of mode i
- $c_i(0)$ initial amount of that mode
- t time measured from zero

Negative rates decay; positive rates grow if the mode is present.



YOUR TURN

Linear Dynamics

Return from modal coordinates

$$V = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \Lambda = \mathrm{diag}(-1, -2)$$

- V eigenvector basis
- Λ matching eigenvalues

For NEW $x_0=(1,1)^T$, which modal description is correct?

A. $c_0=(0,1)^T$; only the -2 mode

B. $c_0=(1,2)^T$; both modes

C. $c_0=(1,1)^T$; both modes

DEFINITION

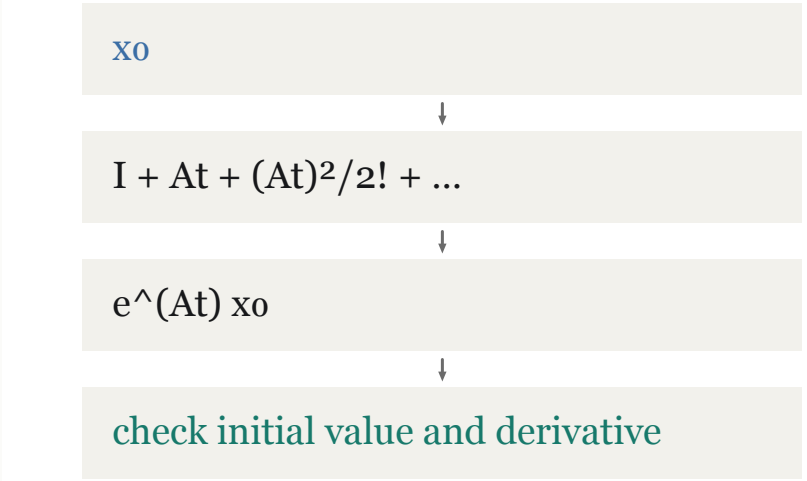
Linear Dynamics

The matrix exponential always exists

$$e^{At} = \sum_{j=0}^{\infty} \frac{(At)^j}{j!}, \qquad x(t) = e^{At} x_0$$

- A any square constant matrix
- j series index; $A^0 = I$
- x_0 initial state

Diagonalization is one way to evaluate the solution, not an existence requirement.



YOUR TURN

Linear Dynamics

Check a candidate for a defective system

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, \quad x_0 = (0, 1)^T, \quad x(t) = (te^{-t}, e^{-t})^T$$

`x(t)` candidate trajectory

`A, x₀` given matrix and start

Does this trajectory solve the given IVP?

A. No: a defective matrix has no solution.

B. No: the factor t forces growth forever.

C. Yes: its derivative and initial state both match.

WORKED EXAMPLE

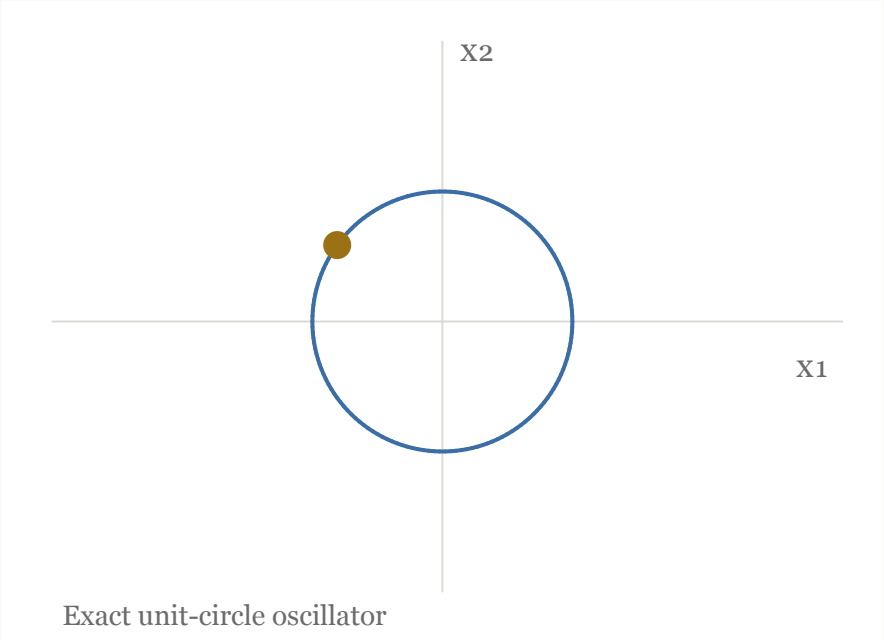
Linear Dynamics

A real rotation has complex eigenvalues

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad x(t) = (\cos t, \sin t)^T$$

- x_0 $(1,0)^T$
- A constant rotation generator
- λ $\pm i$ over the complex field

> This trajectory stays on the unit circle.



Exact unit-circle oscillator

YOUR TURN

Linear Dynamics

Zero real parts need a closer look

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad x_0 = (0, 1)^T$$

- `A` both eigenvalues zero
- `x0` chosen initial state
- `x(t)` state to determine

What does this zero-eigenvalue example do?

A. $x(t)=(0,1)^T$: fixed

B. $x(t)=(t,1)^T$: unbounded

C. $x(t)=(\exp(t),1)^T$: exponential growth

DERIVATION

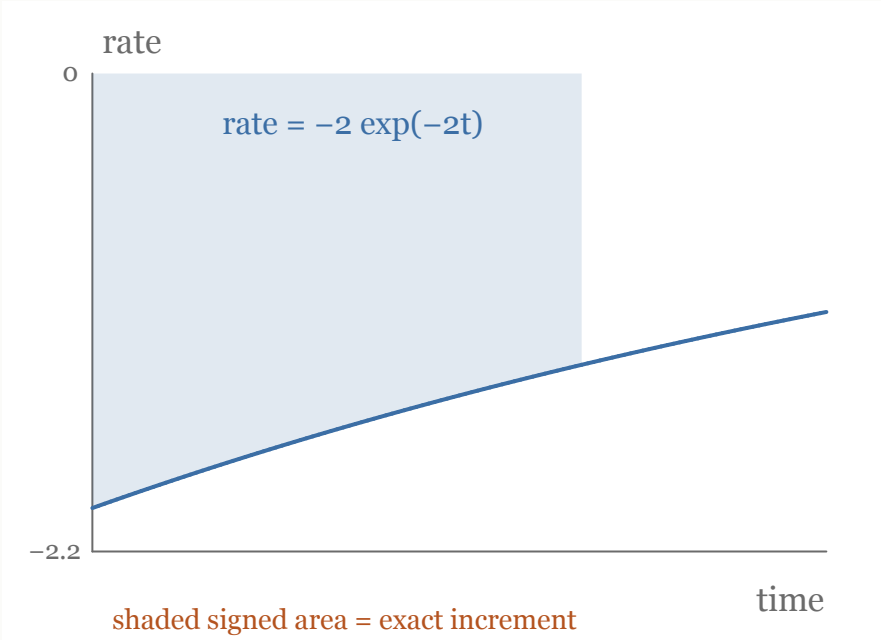
Euler Integration

An exact step is an integral

$$x(t+h) - x(t) = \int_t^{t+h} f(x(s), s)\, ds$$

- `h` positive interval length
- `s` integration variable
- `f(x(s),s)` rate along the unknown path

The fundamental theorem of calculus gives the starting relation.



RULE

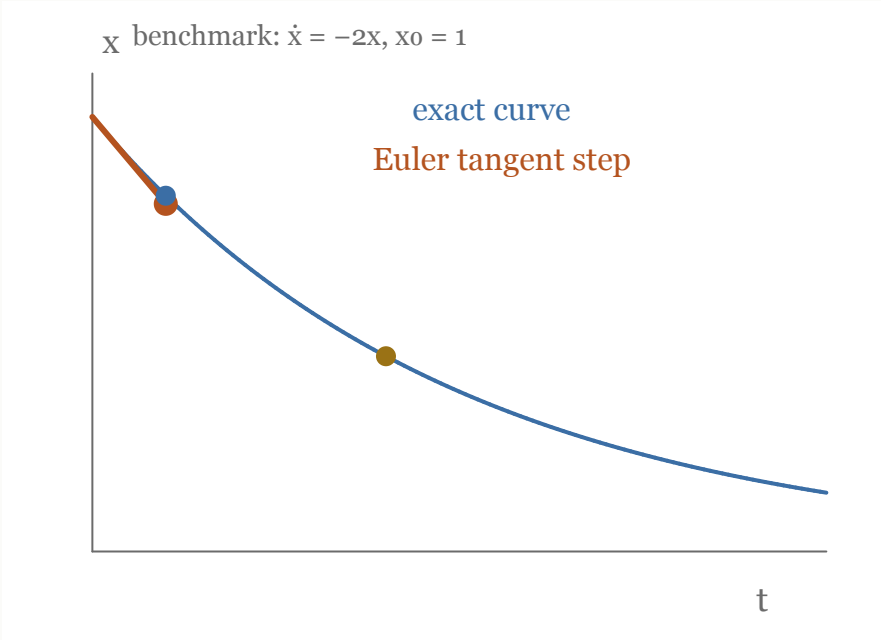
Euler Integration

Euler uses the rate at the start

$$x_{k+1} = x_k + hf(x_k, t_k)$$

- x_k current numerical state
- h positive time step
- $f(x_k, t_k)$ rate at the old state

Every component uses the same old state.



WORKED EXAMPLE

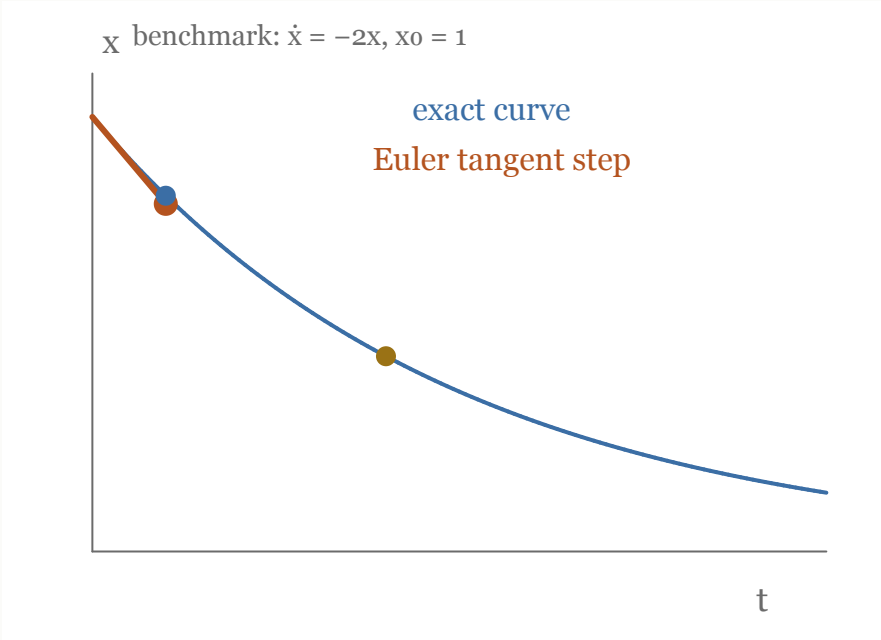
Euler Integration

One step against an exact answer

$$x_1 = 1 + 0.1(-2) = 0.8$$

x_0	one
h	0.1
$f(x)$	$-2x$

> Worked example: exact x(0.1)=0.8187307531; Euler is lower.



YOUR TURN

Euler Integration

Your turn: two larger steps

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.2$$

- h fixed time step
- x_0 initial numerical state
- x_2 approximation at t=0.4

Which Euler pair uses a fresh rate at each step?

A. x1=.6, x2=.36

B. x1=.6, x2=.2

C. x1=.8, x2=.64

Euler Integration

Local error and accumulated error differ

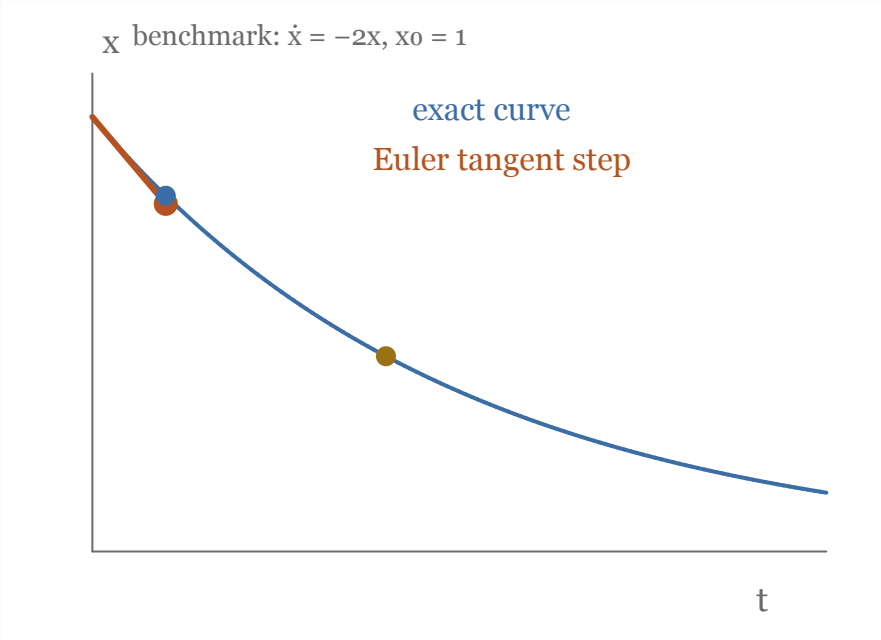
$$x(t+h) = x + h\dot{x} + \tfrac{1}{2}h^2\ddot{x} + O(h^3)$$

$$\ddot{x} = f_t + J_f f$$

`x, f` evaluated on the exact path at time t

f_t, `Jf` partial time derivative and state Jacobian

Assume f is C^2 with bounded derivatives near the finite trajectory.



RECALL AND DERIVATION

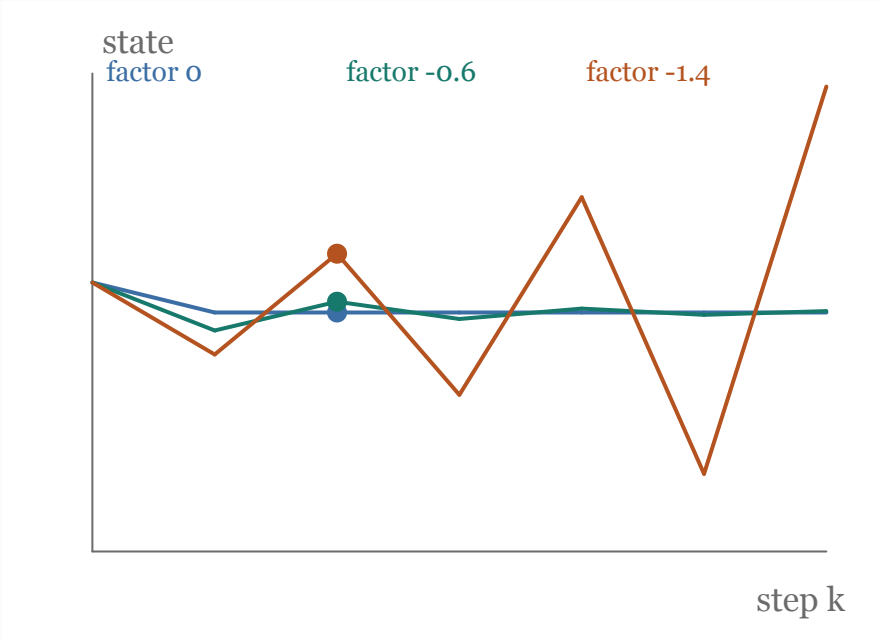
Numerical Stability

A repeated step has a multiplier

$$\dot{x} = \lambda x \quad \Rightarrow \quad x_{k+1} = (1 + h\lambda)x_k$$

- λ constant scalar rate
- h step size
- $1+h\lambda$ numerical amplification factor

Recall Lecture 6: A=−H turns this into the gradient-descent multiplier.



DERIVATION

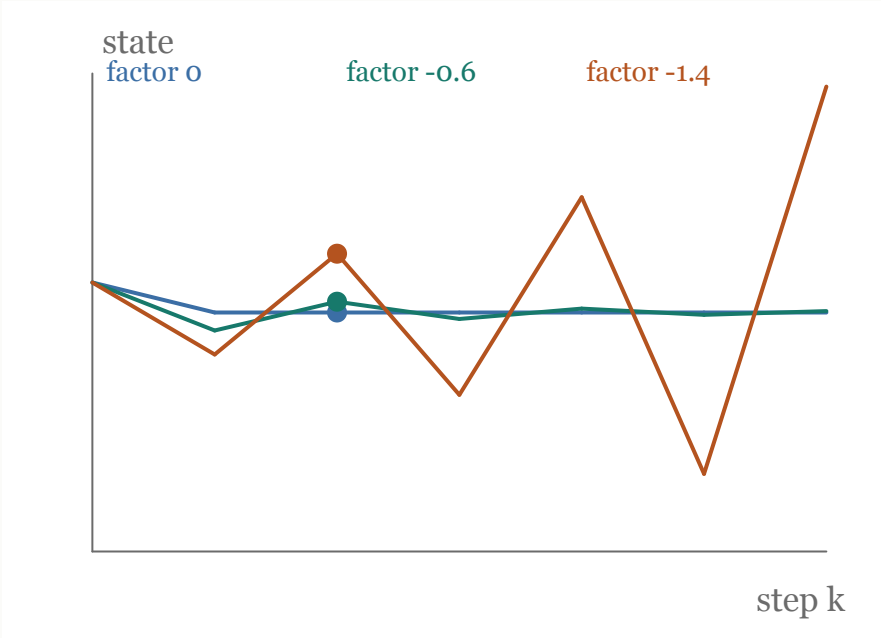
Numerical Stability

The strict inequality decides decay

$$|1 + h\lambda| < 1 \quad \Longleftrightarrow \quad 0 < h < -2/\lambda$$

- λ real negative rate
- h positive step size
- $|1+h\lambda|$ size of each numerical multiplier

At the endpoint the factor is -1: oscillation without decay.



YOUR TURN

Numerical Stability

Predict three step-size regimes

$$\dot{x} = -4x, \quad h \in \{.25, .4, .6\}$$

- λ fixed decay rate
- h candidate steps
- x_0 any nonzero start

For h=.25, .4, .6, which Euler behaviors occur in order?

A. Zero; alternating growth; alternating growth

B. All decay because the ODE is stable

C. Zero; alternating decay; alternating growth

YOUR TURN

Numerical Stability

Every mode must satisfy the bound

$$\dot{x} = Ax, \quad A = \mathrm{diag}(-2, -8)$$

- `A` constant state matrix
- `h` one step used for both modes
- x_0 arbitrary initial state

For Euler, which interval gives decay from EVERY start?

A. $0<h\le 1/4$

B. $0<h<1/4$

C. $0<h<1$

DIAGNOSTIC

Numerical Stability

Euler adds energy to an oscillator

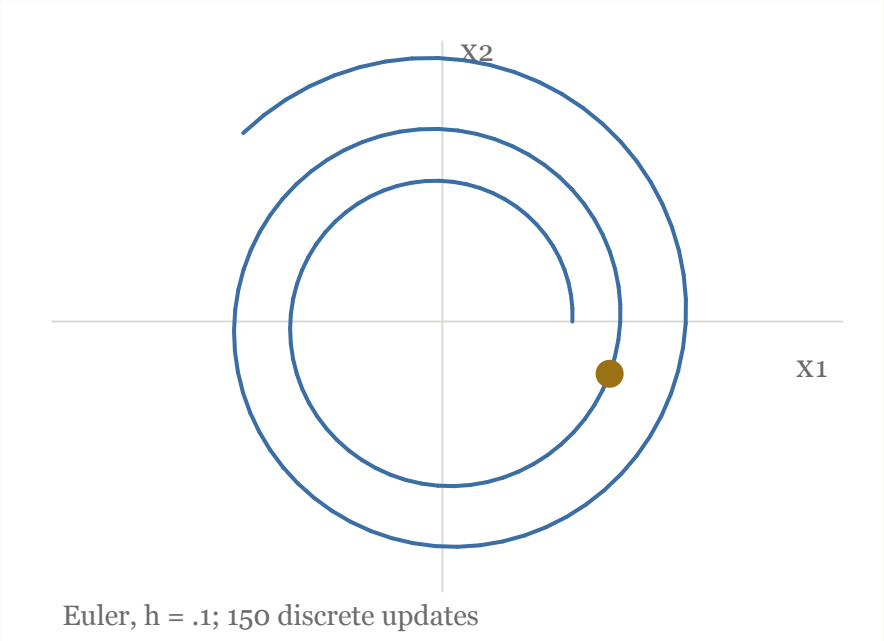
$$\dot{x} = Ax, \quad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$|1 + ih| = \sqrt{1 + h^2} > 1 \quad (h > 0)$$

`±i` eigenvalues of A

`h` positive Euler step

A bounded exact orbit can produce an outward numerical spiral.



Euler, h = .1; 150 discrete updates

RECAP

Numerical Stability

Model validation and solver verification

$$\text{model: } e^{\lambda h} \qquad \text{Euler: } 1 + h\lambda$$

`λ`	continuous rate
`h`	step size
`two factors`	exact and numerical evolution over the same interval

A solver check tests the approximation. Measurements test the physical model.

Same model, two comparisons

Numerical solution

↓ compare with

Exact / refined solution

tests the solver and step size

Model prediction ↔ measurements

tests the physical model

WHY

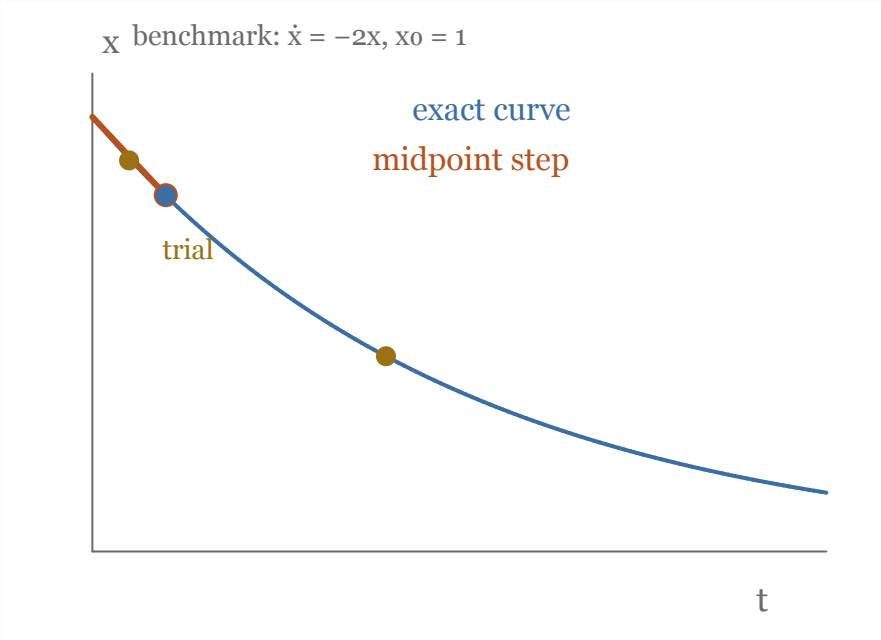
Runge–Kutta Methods

Sample the field inside the step

$$\Delta x = \int_{t_k}^{t_k+h} f(x(s), s)\, ds$$

- Δx exact increment from the current exact state
- f same velocity model
- h one integration interval

A better slope estimate can improve the same time step.



RULE

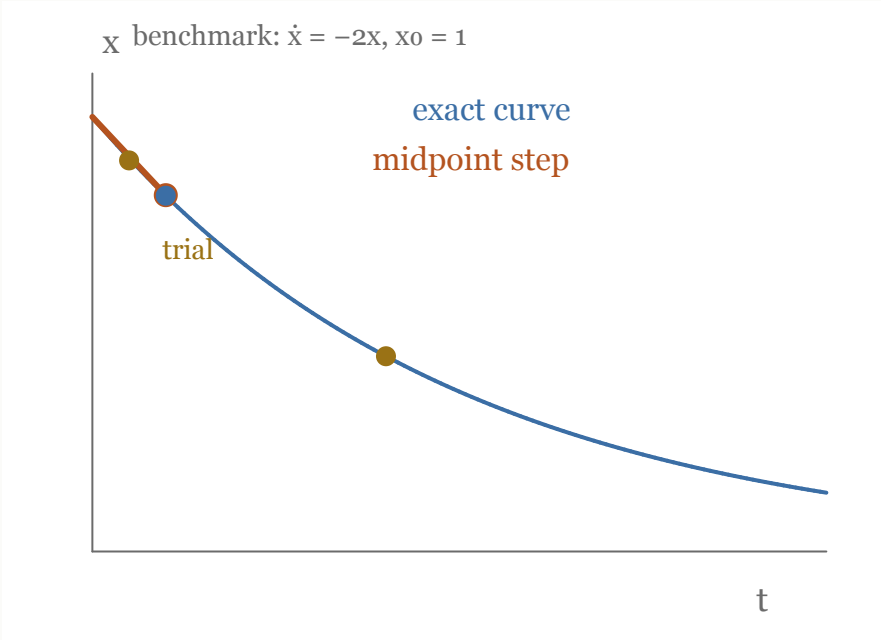
Runge–Kutta Methods

Construct a trial midpoint

$$k_1 = f(x_k, t_k), \qquad k_2 = f(x_k + \tfrac{h}{2}k_1, t_k + \tfrac{h}{2})$$

k_1, k_2	rates, not increments
$h\ k_1/2$	trial half-step displacement
$t_k + h/2$	midpoint time

The first rate predicts a location; the second samples the field there.



WORKED EXAMPLE

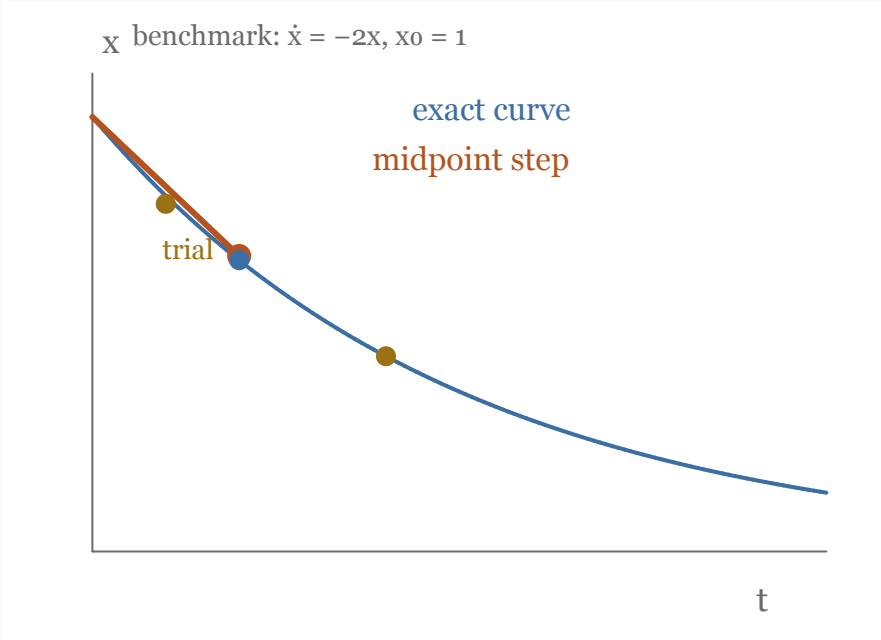
Runge–Kutta Methods

Advance with the midpoint rate

$$x_{k+1} = x_k + hk_2$$

- x_k original current state
- k_2 rate at the trial midpoint
- h full step length

> $\dot{x}=-2x$, $x_0=1$, $h=.2$: $k_1=-2 \rightarrow$ trial $=.8 \rightarrow k_2=-1.6 \rightarrow x_1=.68$.



YOUR TURN

Runge–Kutta Methods

Your turn: halve the midpoint step

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.1$$

`k1,k2` two model evaluations

`h` full step

`x1` accepted next state

Which value is the ACCEPTED midpoint state?

A. .9: the trial midpoint

B. .72: trial state plus full increment

C. .82: original state plus full increment

RULE

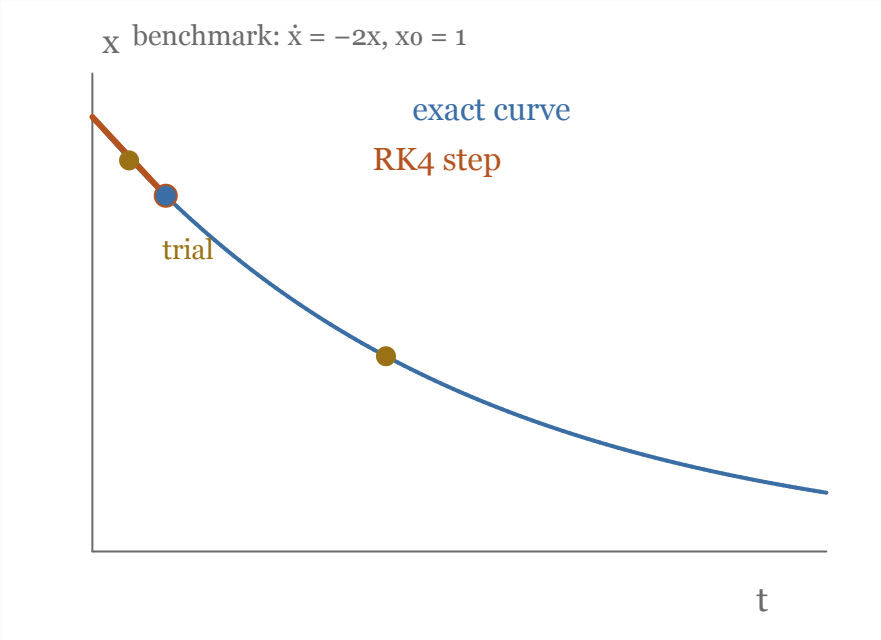
Runge–Kutta Methods

Classical RK4 evaluates four trial rates

$$
\begin{aligned}
k_1 &= f(x, t) \\
k_2 &= f(x + hk_1/2, t + h/2) \\
k_3 &= f(x + hk_2/2, t + h/2) \\
k_4 &= f(x + hk_3, t + h)
\end{aligned}
$$

- `x,t` current state and time
- `h` one full step
- `k₁…k₄` four rates with state/time units

Each trial is anchored at the original x.



WORKED EXAMPLE

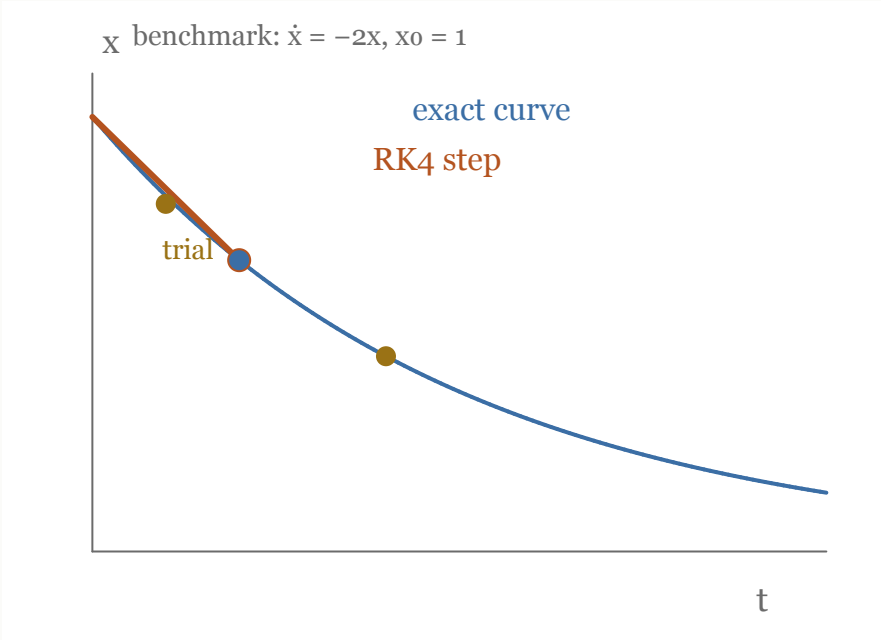
Runge–Kutta Methods

Combine the four rates

$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$k_1 \ldots k_4$	RK4 rates
1,2,2,1	weights before division by six
h	full step length

> $\dot{x}=-2x$, $x_0=1$, $h=.2$: rates $(-2,-1.6,-1.68,-1.328)$ give $x_1=.6704$.



YOUR TURN

Runge–Kutta Methods

Your turn: compare equal model-call budgets

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.1$$

`h` one RK4 step interval

`cost` number of calls to f

How many Euler substeps match ONE RK4 step in model calls?

A. Four, each h/4, over the same interval

B. One, with step h

C. Four, each h, over the same interval

COMPARISON

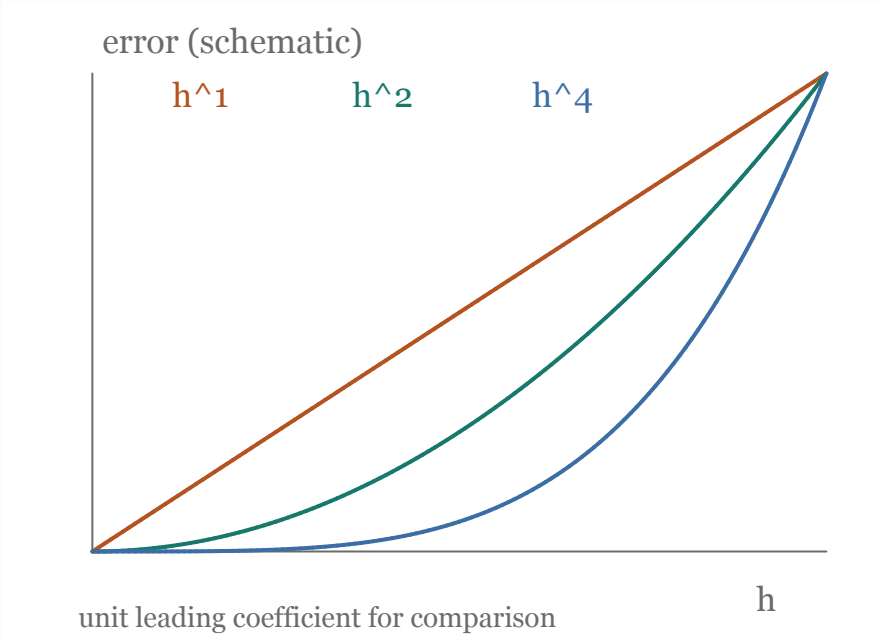
Runge–Kutta Methods

Accuracy improves at a computational price

$$\varepsilon_{\text{Euler}} = O(h), \quad \varepsilon_{\text{midpoint}} = O(h^2), \quad \varepsilon_{\text{RK4}} = O(h^4)$$

- ε norm of state error at a fixed final time
- h step tending to zero
- assumptions smooth, well-posed finite-interval problem

Halve h and check convergence; RK4 is not unconditionally stable.



MODEL

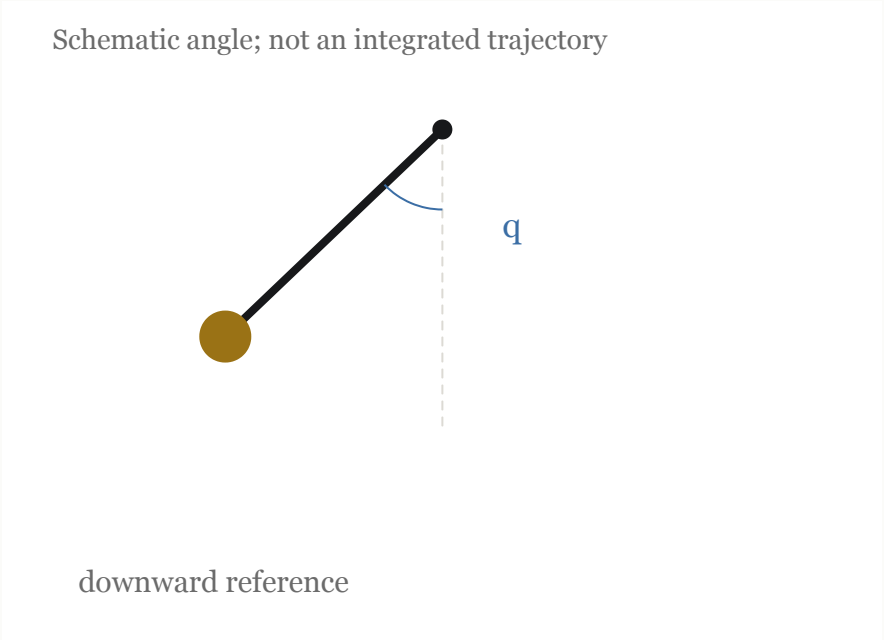
Pendulum Simulation

The restoring acceleration has a sign

$$\dot{q} = v, \qquad \dot{v} = -\frac{g_0}{\ell} \sin q$$

- `q` angle from downward vertical
- `v` angular velocity
- `g₀, ℓ` positive gravity acceleration and length

Unforced, undamped motion is the model for this comparison.



DEFINITION

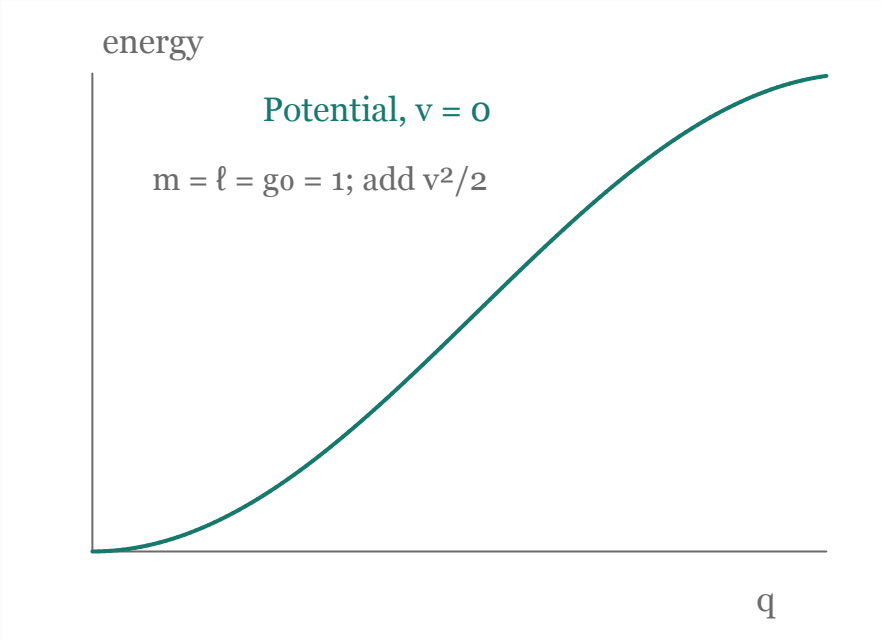
Pendulum Simulation

Mechanical energy is a diagnostic

$$E(q, v) = \tfrac{1}{2} m \ell^2 v^2 + m g_0 \ell (1 - \cos q)$$

`q,v`	angle and angular velocity
`m,ℓ,g₀`	mass, length and gravity
`two terms`	kinetic and potential energy

The zero of potential energy is a convention; energy differences have meaning.



DERIVATION

Pendulum Simulation

The exact dynamics conserve energy

$$E = \tfrac{1}{2} m \ell^2 v^2 + m g_0 \ell (1 - \cos q)$$

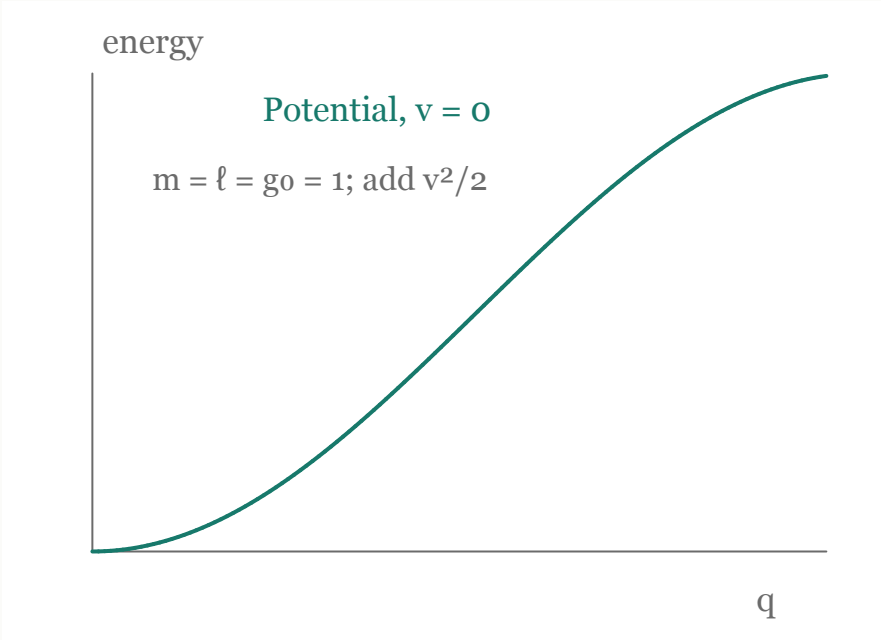
$$\dot{E} = m \ell^2 v \dot{v} + m g_0 \ell \sin q \, \dot{q} = 0$$

$\dot{q}=v$ kinematic identity

$\dot{v}$ $-(g_0/\ell)\sin q$

$\dot{E}$ rate of mechanical energy

The two terms cancel only for the stated unforced, undamped model.



YOUR TURN

Integrated Pendulum Problem

Choose the state and the check first

$$\ddot{q} = -\sin q, \quad q_0 = 1, \quad v_0 = 0$$

`q₀,v₀` angle and angular velocity

`model` unforced, undamped; m=ℓ=g0=1

Which state and field close this second-order model?

A. x=q; f=−sin q

B. x=(q,v)ᵀ; f=(v,−sin q)ᵀ

C. x=(q,v)ᵀ; f=(−sin q,v)ᵀ

YOUR TURN

Integrated Pendulum Problem

Compare the first accepted states

$$x_0 = (1, 0)^T,\ h = .1,\ f(q, v) = (v, -\sin q)^T$$

`x=(q,v)ᵀ` angle and angular velocity

`h` full time step

Why can midpoint change q when v0=0?

A. Nonzero trial velocity.

B. Twice the physical time.

C. A different initial velocity.

YOUR TURN

Integrated Pendulum Problem

Repeat to the same final time

$$\dot{x} = (v, -\sin q)^T,\ x_0 = (1, 0)^T,\ T = 10$$

- `N` step count
- `E` v²/2+1−cos q
- `ΔE` E at T minus E at 0

Halve h: how do we keep the endpoint fixed?

A. Keep N.

B. Stop at equal E.

C. Double N.

YOUR TURN

Integrated Pendulum Problem

Compare two energy-matched states

$$E(q, v) = v^2/2 + 1 - \cos q$$

`q,v` angle and angular velocity

`E` energy

What does equal energy of two pendulum states establish?

A. Their angles and velocities are equal.

B. They can still be different states.

C. Both numerical trajectories are exact.

YOUR TURN

Independent Transfer

Transfer: a moving pendulum near the top

$$q_0 = \pi, \quad v_0 = 1, \quad h = .1, \quad m = \ell = g_0 = 1$$

`q₀,v₀`	initial angle and angular velocity
`f(q,v)`	$(v, -\sin q)^T$
`E(q,v)`	$v^2/2 + 1 - \cos q$

One call: which method and energy claim is correct?

A. Euler; this step decreases energy.

B. Midpoint; one call is enough.

C. Euler; every step increases energy.

YOUR TURN

Simulation Checklist

Choose a complete simulation workflow

$$\dot{x} = f(x, t), \qquad x(t_0) = x_0$$

- `x` complete state
- `f` model rate
- `h` chosen integration step

For this pendulum simulation, which workflow distinguishes errors?

A. Use q alone; shrink h until the curve looks smooth.

B. Use f as the next state; inspect only energy.

C. Use (q,v), integrate rates, compare states and energy.

CONNECTION

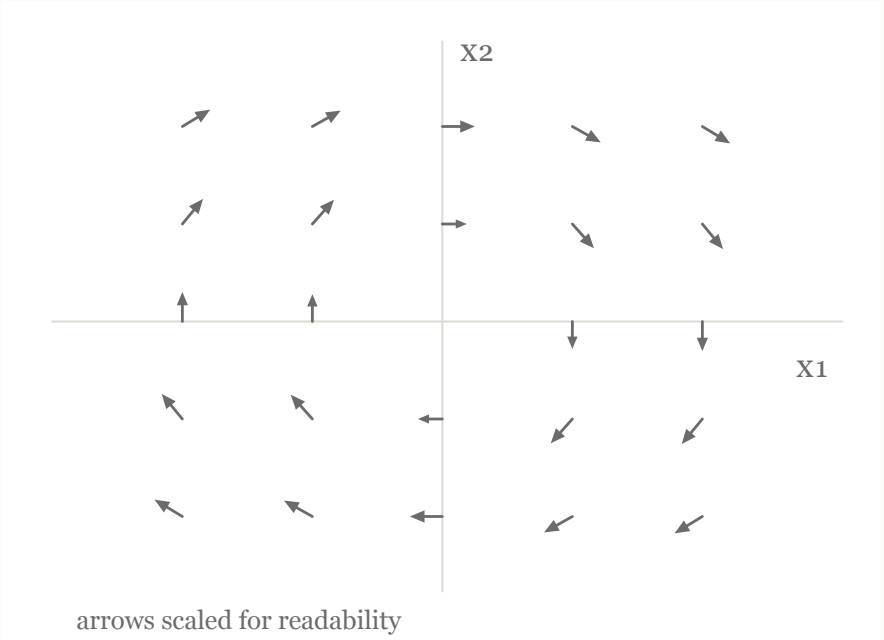
Simulation Checklist

A learned field still needs an integrator

$$\dot{x} = v_\theta(x, t)$$

- `vθ` a field represented by a learned model
- `x` generated state
- `t` continuous evolution coordinate

A learned field supplies rates; Euler or RK advances the state.



References

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References

Sources for integration and stability

1. Kværnø, A. Numerical Solution of ODEs, part 2. NTNU, 2022.

2. Driscoll, T. & Braun, R. Fundamentals of Numerical Computation, §6.4. Online textbook.